Serre's Theorem

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Acknowledgements

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The Cayley Graph

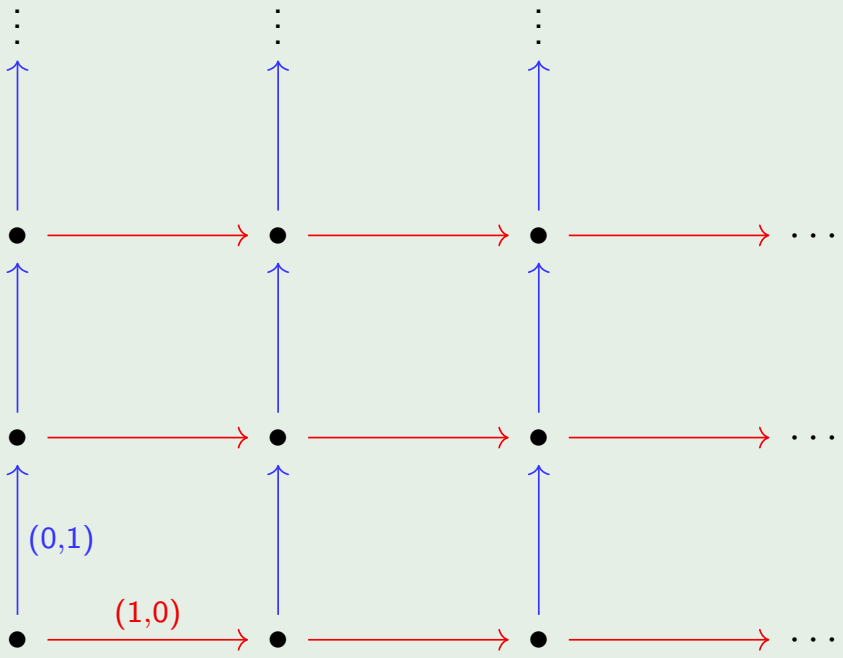
Definition

Let G be a group generated by a set S. The **Cayley graph of G with respect to S** is the graph $\Gamma(G, S)$, defined as follows: The vertex set is G. Each $s \in S$ is assigned a colour c_s. Then for each g, there is a c_s-coloured edge from g to sg.

The Cayley Graph

Examples

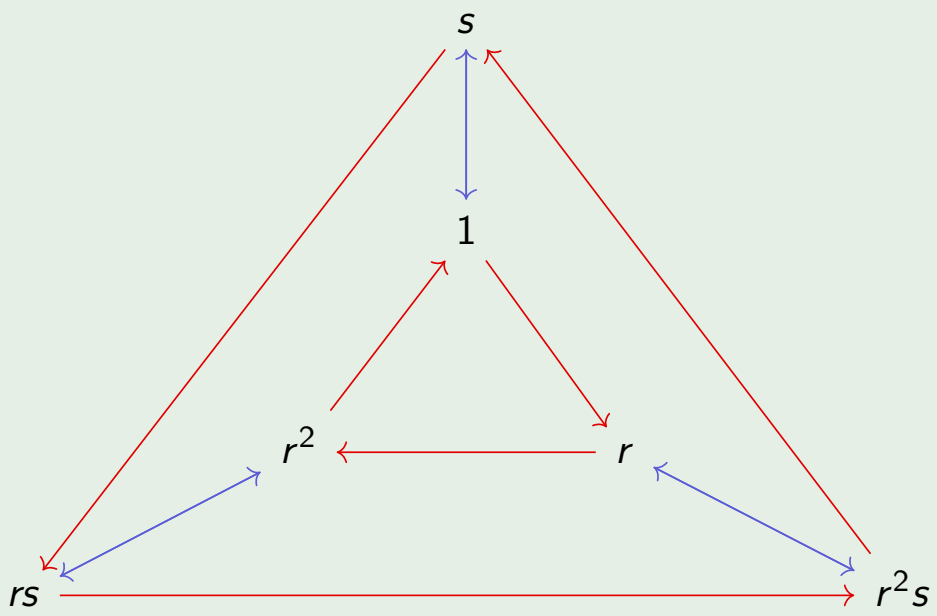
Let $G = \mathbb{Z}^2$, generated by $\{(0,1),(1,0)\}$. The Cayley graph is an integer lattice:



The Cayley Graph

Examples

Let $G = D_3$, generated by rotation r and reflection s. We get this:



Free Groups

Definition

A group with n generators and no relations between them is called a **free group**. Such groups are denoted as F_n.

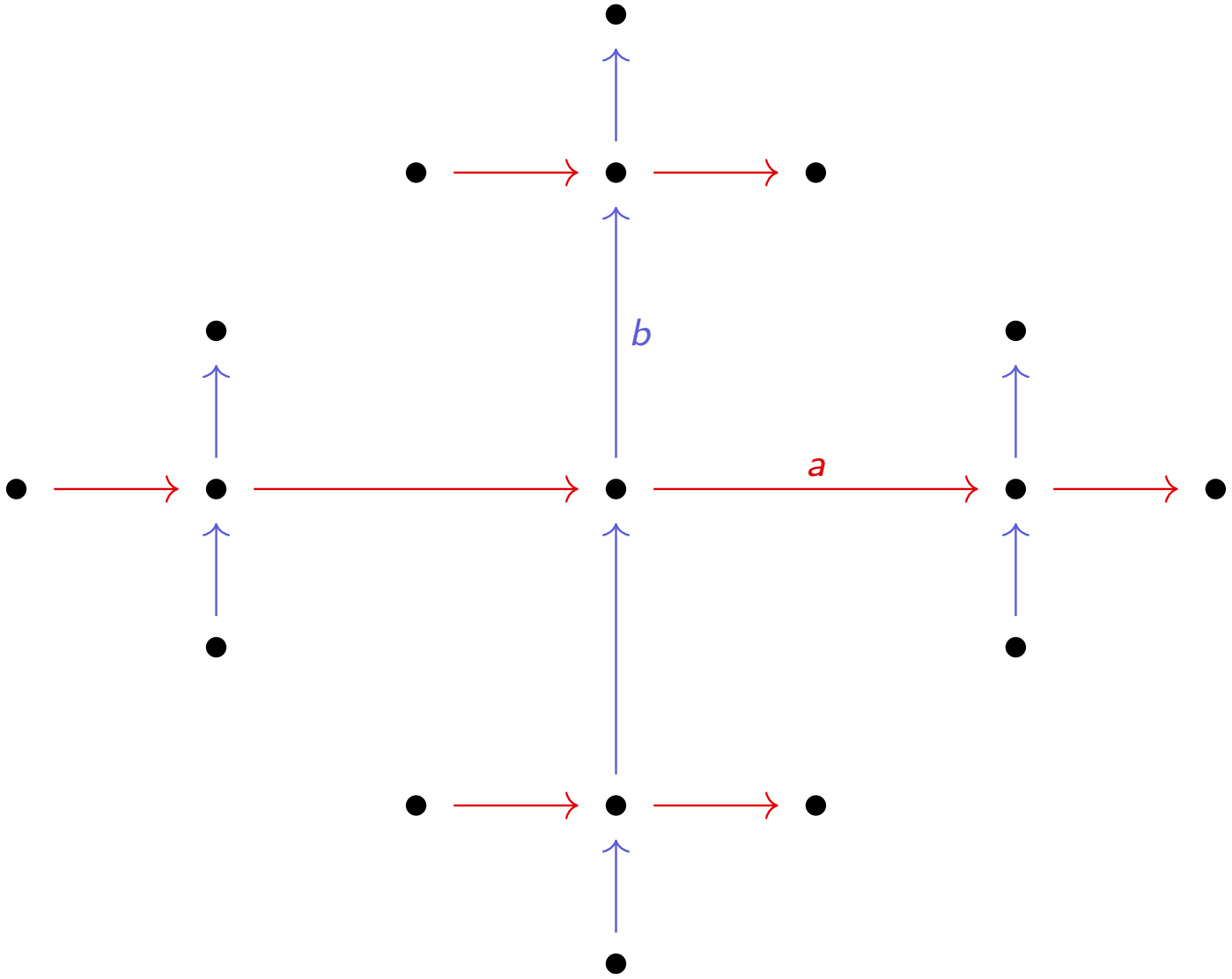
Examples

In $SL_2(\mathbb{Z})$ (2×2 integer matrices of determinant 1), the group generated by

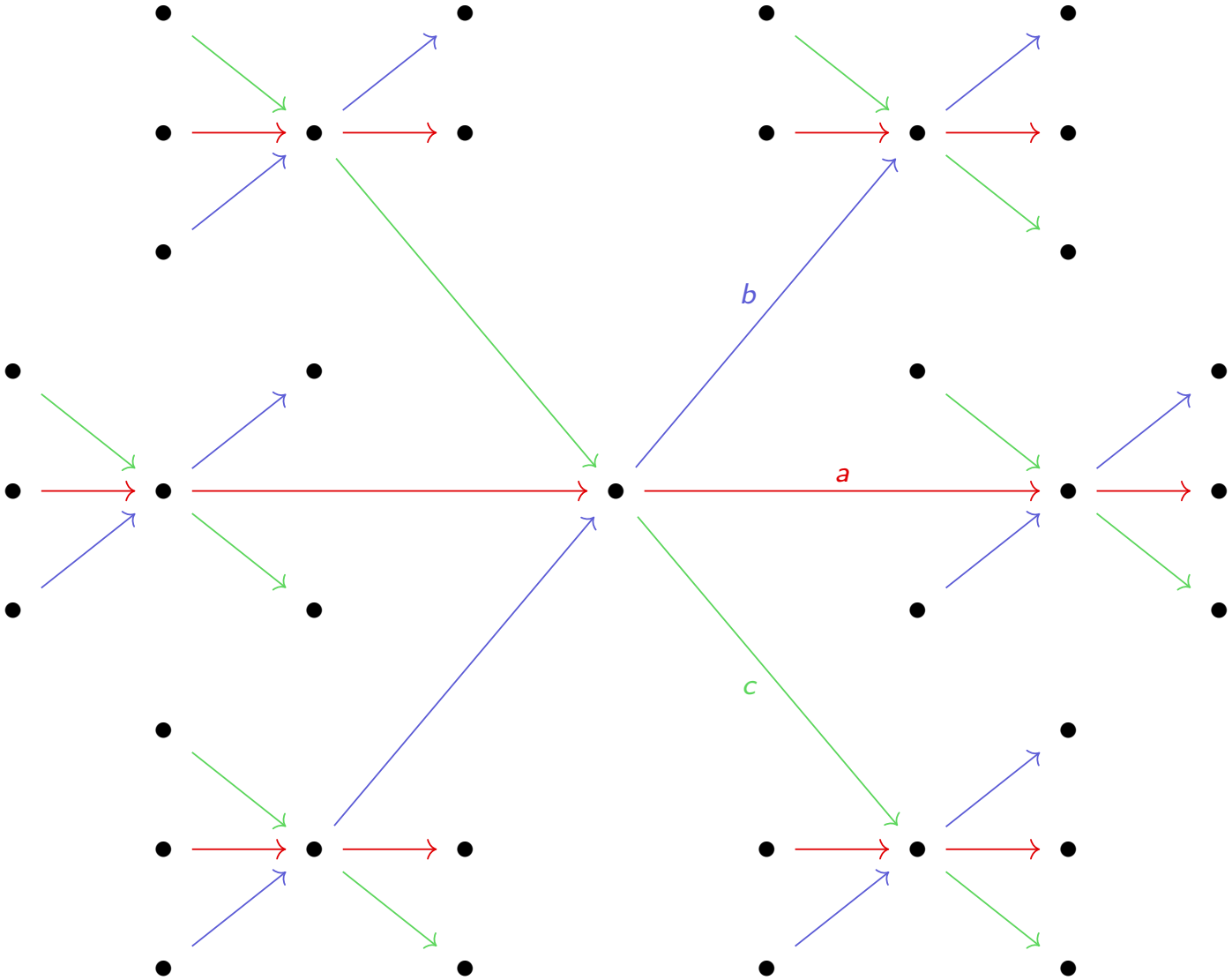
$$A = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

is a free group of rank 2.

The Cayley Graph of $F_2 = \langle a, b \rangle$



The Cayley Graph of $F_3 = \langle a, b, c \rangle$

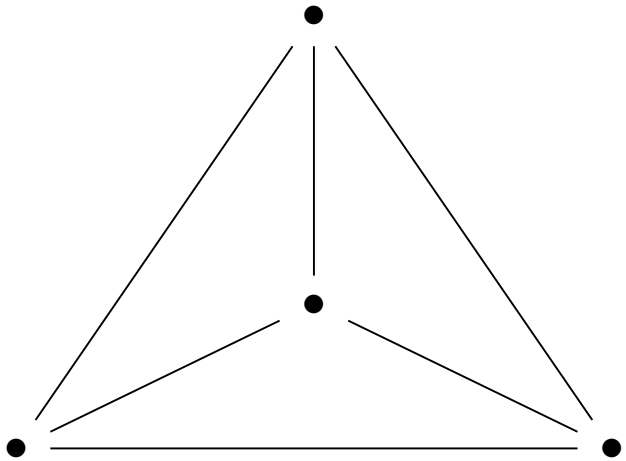


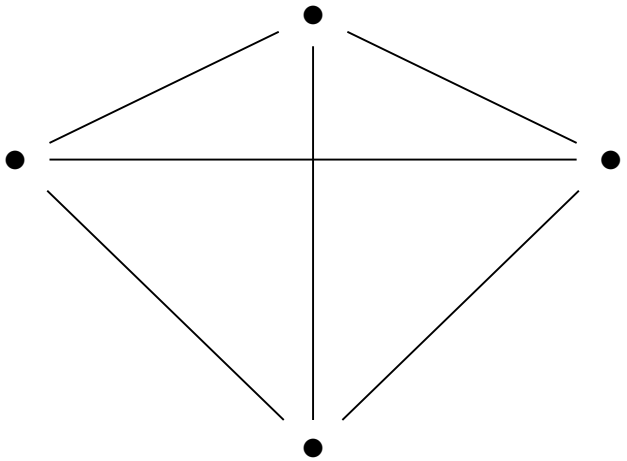
Graphs and Graph Isomorphisms

Definition

We say that two graphs $\Gamma_1 = (V_1, E_1), \Gamma_2 = (V_2, E_2)$ are **isomorphic** if and only if there is a bijection $f : V_1 \to V_2$ so that

$$(v, w) \in E_1 \iff (f(v), f(w)) \in E_2$$



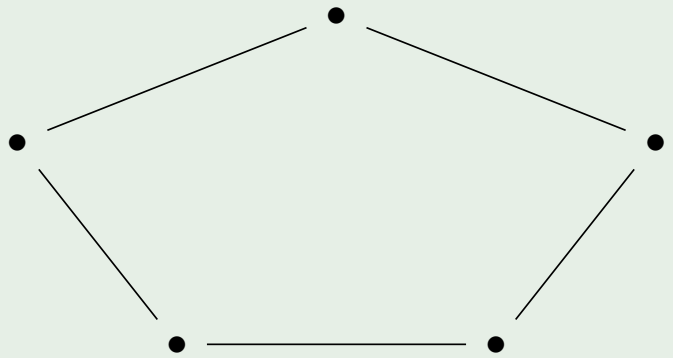


Graph Automorphisms

A graph **automorphism** is an isomorphism from a graph Γ to itself. The set of automorphisms of Γ is a group, denoted $Aut(\Gamma)$.

Examples

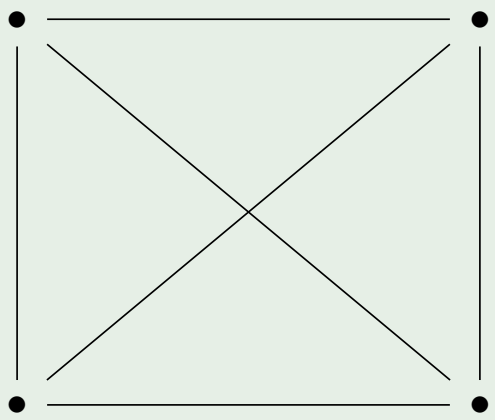
If Γ is the graph of a regular pentagon, then $Aut(\Gamma) = D_5$.



Graph Automorphisms

Examples

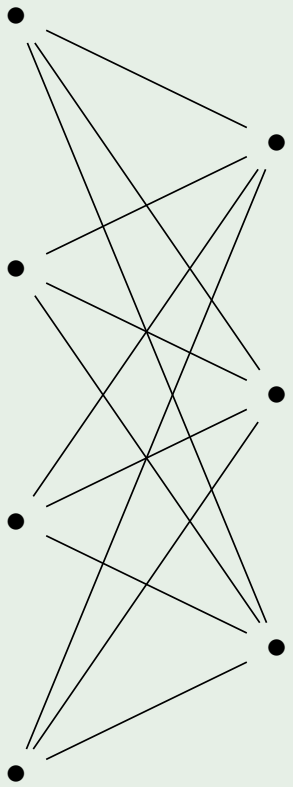
If $\Gamma = K_4$, then $Aut(\Gamma) = S_4$.



Graph Automorphisms

Examples

If $\Gamma = K_{4,3}$, then $Aut(\Gamma) = S_4 \times S_3$.



Acting on a Graph

Definition

Let G be a group and Γ a graph. A homomorphism $G \to Aut(\Gamma)$ is called an **action** of G on Γ, and we say that G **acts** on Γ.

Examples

1. D_5 acts on the graphs of regular pentagon by rotation and reflection.
2. S_4 acts on K_4 by permuting the vertices.
3. $S_4 \times S_3$ acts on $K_{4,3}$ by permuting the vertex sets.

Acting Freely

Definition

Let G act on Γ. We say that the group action is **free** if for any vertex v, $g \cdot v = v$ if and only if $g = 1$. We say that G acts **without inversion** if for any edge $(v, w) \in E$ and $g \in G$, either $g \cdot v \neq w$ or $g \cdot w \neq v$.

Examples

- S_4 acts freely on K_4, but with inversion.
- $\mathbb{Z}_6$ acts without inversion on the graph of a triangle, but not freely.
- $\mathbb{Z}^2$ acts freely on its Cayley graph without inversion.

Barycentric Subdivision

Any action on a graph may be made into one without inversion by passing to the graph's **barycentric subdivision**:

Acting on a Cayley Graph

Lemma

Every group acts freely on its Cayley graph.

Proof.

G acts on its Cayley Graph by left multiplication on vertices. If $g \cdot v = v$ for all vertices v, taking $v = 1$ gives $g \cdot 1 = 1$, hence $g = 1$. $\square$

Serre's Theorem: Part 1

Theorem (Serre's Theorem: Part 1)

If G is a free group, then there is a tree T on which G acts freely, without inversion.

Table of Contents

1. Cayley Graphs

2. Tree Tilings

Tilings

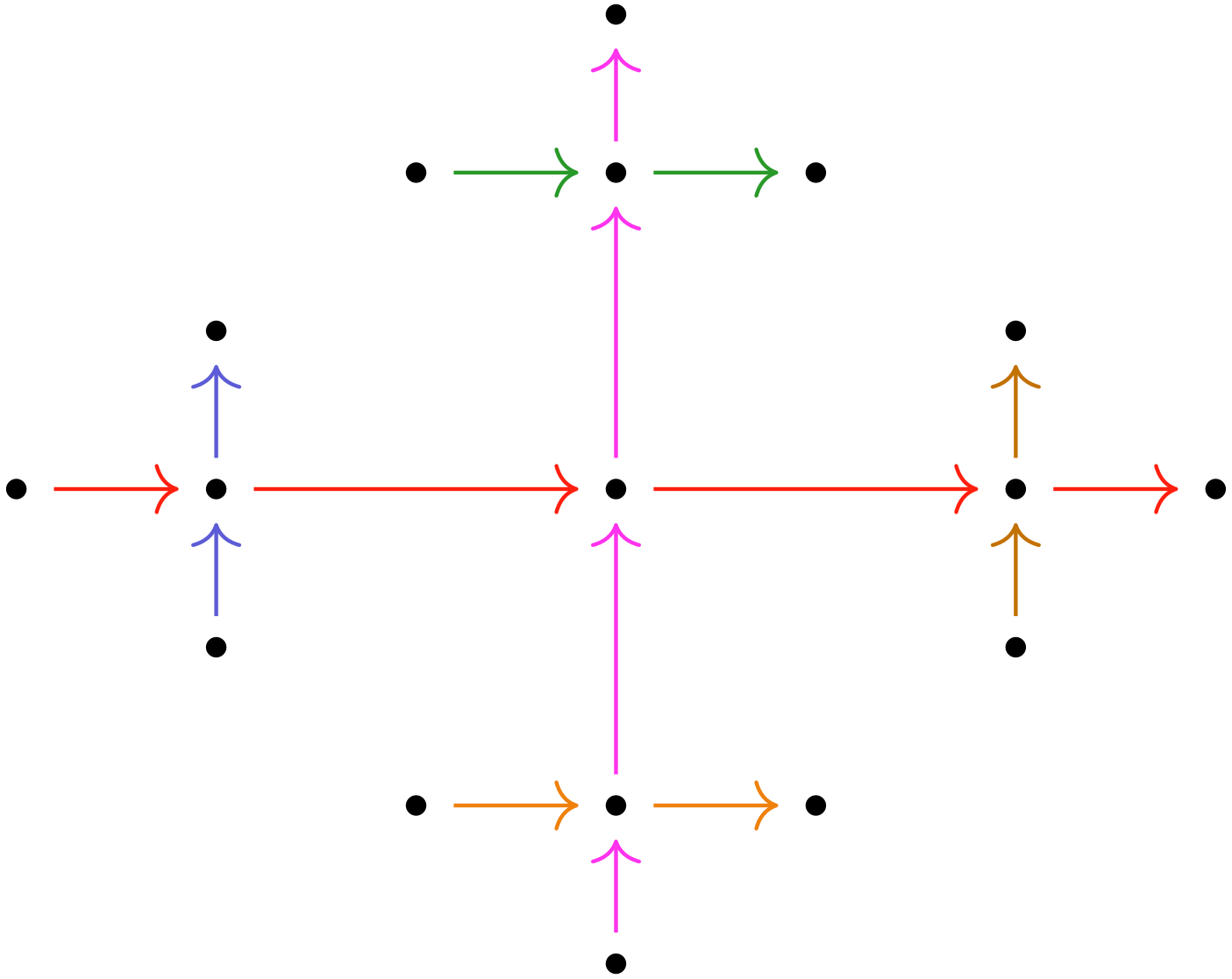
Definition

A **tile** of a tree T is any subtree of T.

A **tiling** of T is a collection of tiles of T such that

1. The union of all tiles is T, and
2. Two tiles intersect, at most, at a vertex.

A Tiling of F_2's Cayley Graph

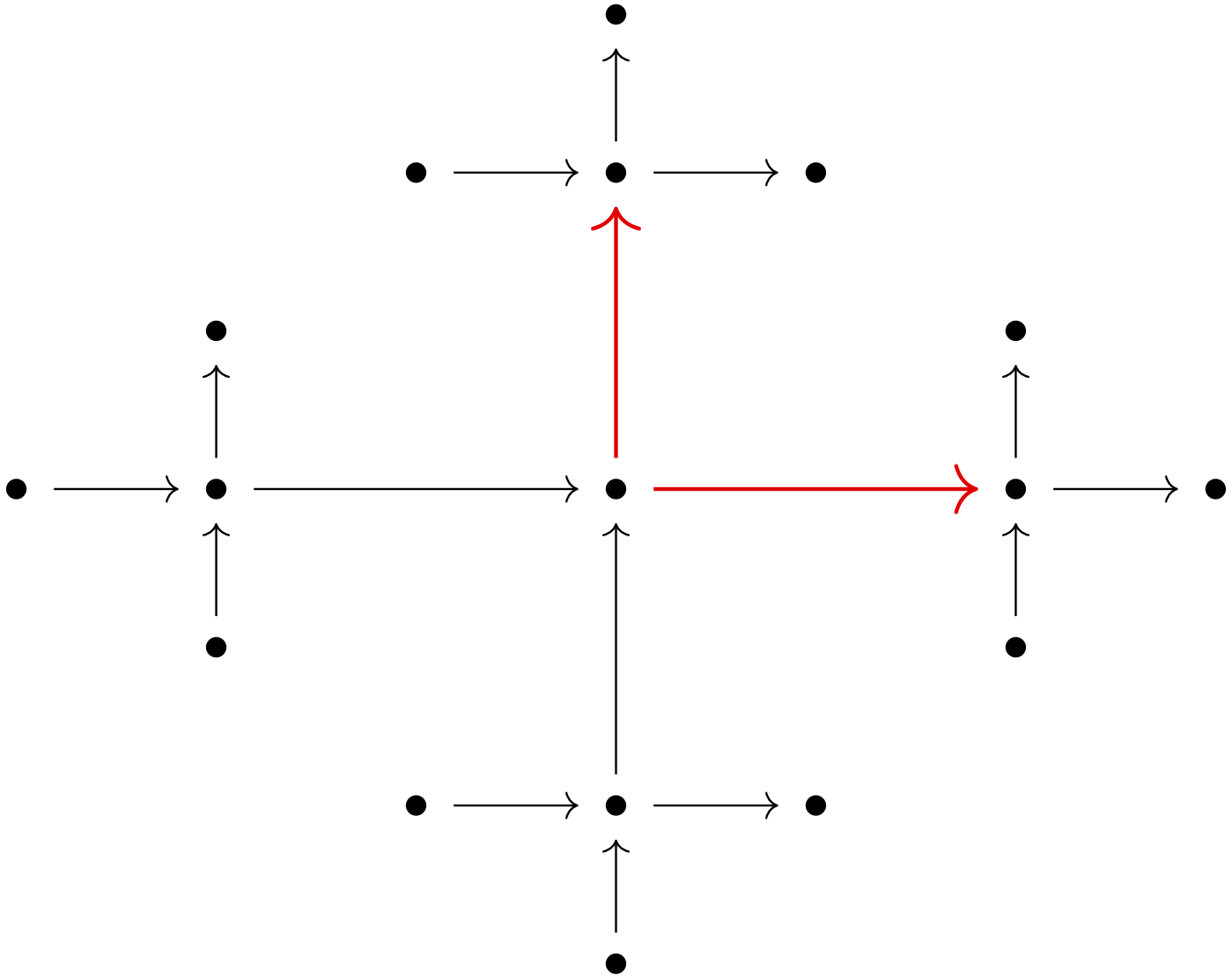


G-Tilings

Definition

Let G act on T. If there is a tiling of T such that there is a tile T_1 so that the set of all tiles is $\{gT_1 : g \in G\}$, then we call the tiling a G-**tiling**.

A G-Tiling of F_2's Cayley Graph



G-Tilings

Theorem

If G acts on T freely and without inversion, then there is a G-tiling of T.

G-Tilings

To construct our G-tiling, fix a vertex v_0 in T's barycentric subdivision (which we will from now on just call T), and consider its orbit

$$\mathcal{O}(v_0) = \{gv_0 : g \in G\}$$

The tile T_g is then defined by

$$V(T_g) = \{v \in V(T') : d(v, gv_0) \leq d(v, g'v_0) \, \forall g' \neq g\}$$
$$E(T_g) = \{(u, v) \in E(T') : u, v \in V(T_g)\}$$

G-Tilings

The T_g's form a G-Tiling; in fact, every $g \in G$ corresponds to exactly one tile T_g and vice versa. Let

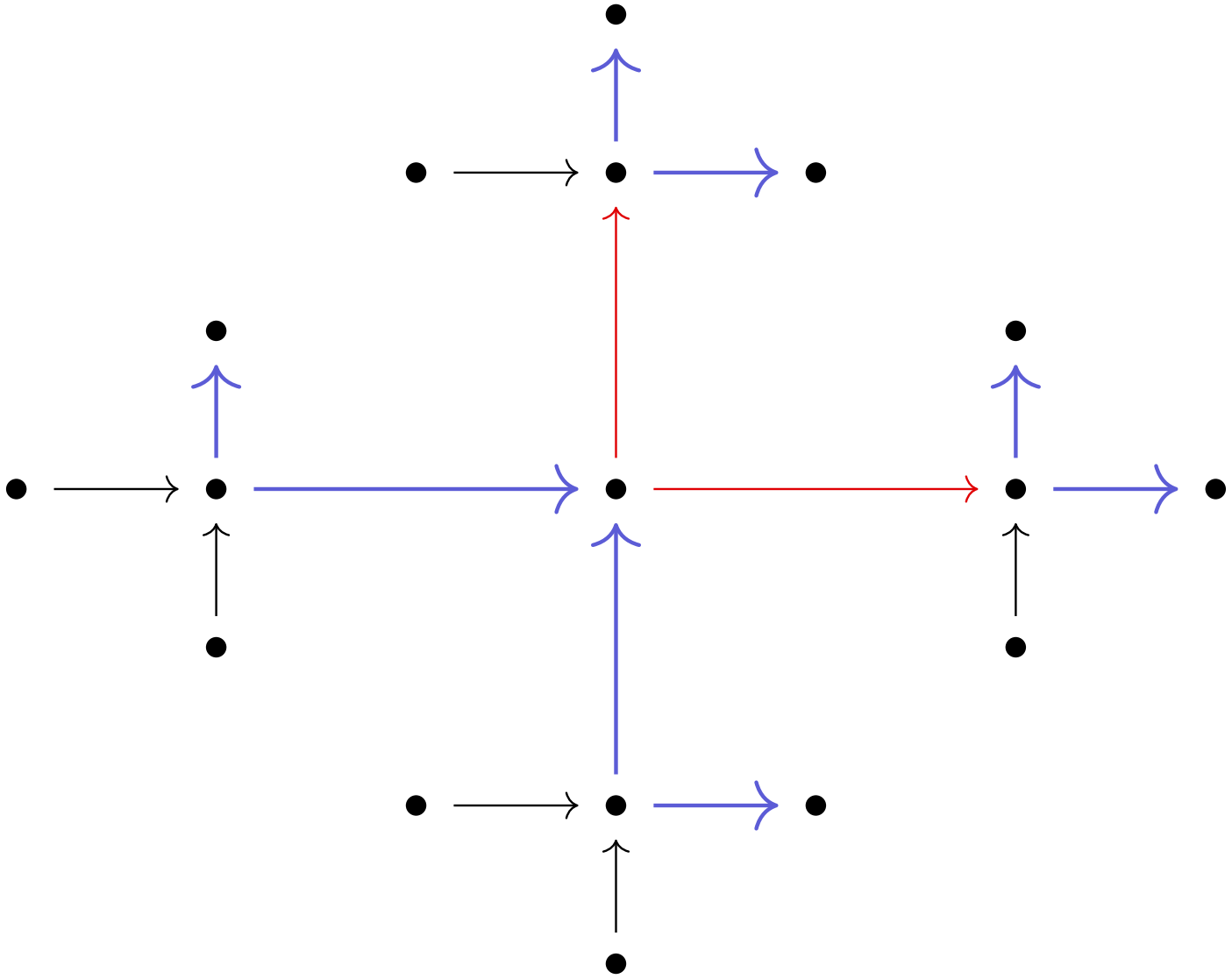
$$S = \{g \in G \| T_g \cap T_1| = 1\}$$

be the elements of G whose corresponding tiles intersect with T_1 at a vertex. This is the set of directions by which we can translate T_1.

Lemma

S is generating set for G.

$S = \{a, b, a^{-1}, b^{-1}\}$



Serre's Theorem: Part 2

Theorem (Serre's Theorem: Part 2)

If G acts freely on a tree T, without inversion, then G is a free group.

Serre's Theorem

Combining Part 1 and Part 2:

Theorem (Serre's Theorem (1977))

G is a free group if and only if it acts freely on a tree without inversion.

The Nielsen-Schreier Theorem

Corollary (Nielsen-Schreier Theorem)

Every subgroup of a free group is itself free.

Proof.

An action of a group G on a tree trivially restricts to an action of $H \leq G$ on the tree. As $1 \in H$, if the action of G is free, so too must the action of H. $\square$